

MX17 Detector 3 — June 2026 Cosmic-Bench Deep Dive:

micro-TPC angle correlation, drift-velocity measurement,
drift-gap determination, and gas diagnosis

Cosmic-bench analysis, Saclay

runs `mx17_det3_saturday_scan_6-27-26` and `mx17_det3_p2_det1_overnight_6-27-26`

2 July 2026 — rev. 3 July 2026 (drift gap & gas sections)

Abstract

We analyse the final det3 weekend campaign: a 2.4 h run at the operating point (resist 490 V, drift 1000 V; configured for 12 h but stopped early for the overnight p2 run), a six-point drift-HV scan (100–1100 V), and two resist-HV scans. Four results: (i) the universally observed *off-diagonal* micro-TPC angle correlation is quantitatively explained as an additive charge-spreading bias, $\tan \theta_{\text{det}} \simeq \tan \theta_{\text{ref}} \pm w/z_{\text{vis}}$ with $w/z_{\text{vis}} = 0.115$ — a detector constant, not a rotation and not an analysis bug; (ii) exploiting this understanding, a bias-free *ridge-fit* estimator measures the electron drift velocity to 1–3 % statistical precision per 15-minute scan point, giving $v_d(1000 \text{ V}) = 28.1(7) \mu\text{m/ns}$ and the full $v_d(E)$ curve; (iii) the drift gap is mechanically $\approx 30 \text{ mm}$, but only the 19.4 mm of drift column closest to the mesh produces recorded signal: the strip arrival-time distribution carries an exponential tail beyond the apparent edge that terminates at the 30 mm drift time, and the strip amplitude decays with drift *depth* — electrons from the far third of the gap are lost to **attachment** in the contaminated gas; (iv) the measured $v_d(E)$ is far below the Magboltz prediction for clean Ar/iC₄H₁₀ 95/5; with the 30 mm field scale it is reproduced by Ar/iso 95/5 with 1–1.5 % water, and the measured attachment length matches 0.2–0.4 % O₂ — one air/moisture contamination explains both anomalies. A wrong bottle of Ar/CO₂ 90/10, though matching $v_d(E)$ closely, is excluded by the amplification (+50 V at iso-gain) and by its zero attachment.

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1 Datasets, alignment and headline performance

1.1 Runs analysed

All data were taken with det3 (mx17_3, bulked June 15) in the *top* bench slot: FEU 7 = X strips, FEU 8 = Y strips, nominal $z = 702$ mm, strip pitch 0.78 mm, gas nominally Ar/iC₄H₁₀ 95/5 at atmospheric pressure (Saclay, ≈ 745.8 Torr), non-zero-suppressed DREAM readout (32 samples $\times$ 60 ns = 1.92 μ s window).

subrun	purpose	dur.	decoded statistics
long_run_resist_490V_drift_1000V	operating point	141 min	complete (4 files): 29k events, 47k M3 tracks
drift_scan_...drift_{100..1100}V	drift scan, 6 pts	15 min	~ 5 k M3 tracks / point
hv_scan_resist_{425..525}V	resist scan 1, 11 pts	15 min	~ 2 k clean rays / point
hv_scan2_resist_{460..520}V	resist scan 2, 7 pts	15 min	~ 2 k clean rays / point
long_run_p2_det1_sanity_check	overnight (p2 run)	—	53k rays; current headline det3

Table 1: Weekend det3 datasets. The Saturday long run was configured for 720 min but was stopped after 141 min (trigger-timestamp span; the p2 overnight run started at 01:33 on the same bench) — its 4 file pairs are the complete dataset.

1.2 Alignment

The long run is aligned to the M3 reference tracker with the standard iterative z /rotation/translation procedure (spark veto at 50 strips, clean-cluster subset for the geometry fit). Both weekend runs converge to the *same* geometry — $z = 714$ mm, $\theta = 89.45^\circ$, translation offsets agreeing to 10 μ m — confirming the detector did not move between runs. The long-run alignment is therefore used as the seed for every scan point, with only a translation refit per point (observed shifts < 0.1 mm).

1.3 Headline performance

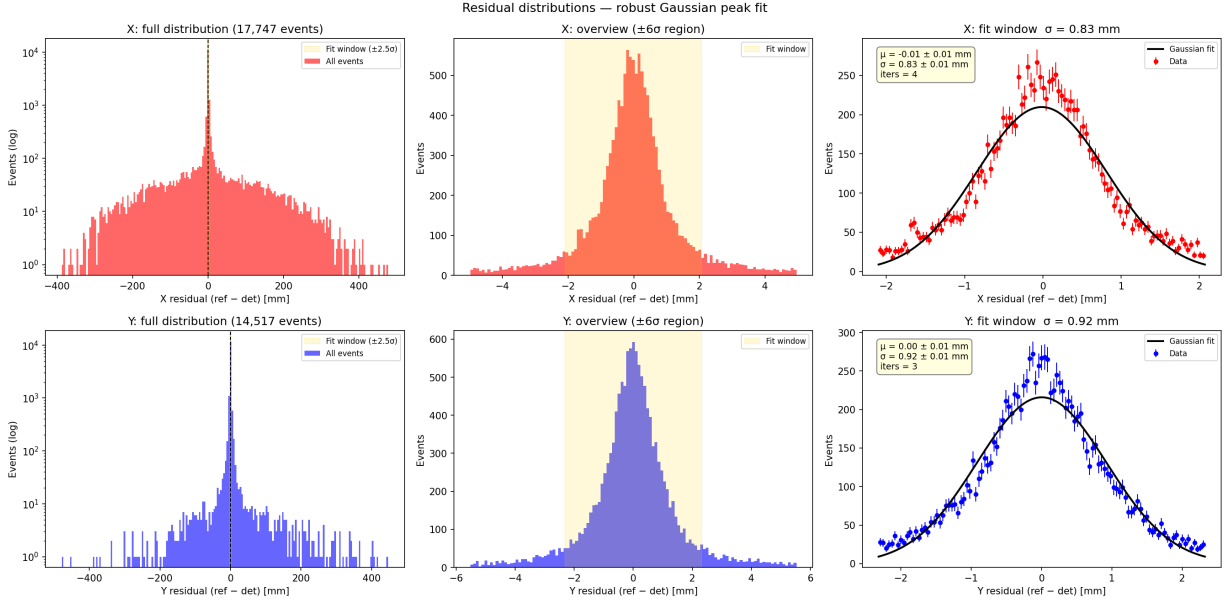


Figure 1: Saturday long run: residuals between the aligned detector hit and the projected M3 track. The Gaussian core gives $\sigma_x = 0.83$ mm, $\sigma_y = 0.92$ mm.

From the efficiency breakdown of the Saturday long run (fixed active box, clean M3 single tracks, $R = 5$ mm matching): the detector fires on 97.6 % of through-going muons, reconstructs an X+Y point for 80.2 %, and 65.8 % land within 5 mm of the track (a 14.4 % *reco-far* tail). The higher-statistics p2 overnight run gives the June-headline 79.7 % within 5 mm with a much smaller tail; the difference (larger far tail in the Saturday run) correlates with the higher spark fraction at resist 490 V (Sec. 7) and can be probed further with the p2 dataset. Either way det3 remains the best detector of the June batch, at sub-mm core resolution in both slots and both weekends.

2 The micro-TPC measurement

2.1 Principle

A single conversion/drift gap of depth g sits above the micromesh; the avalanche charge is read out on strips of pitch $p = 0.78$ mm (Fig. 2). A muon crossing at angle θ (measured from the detector normal, i.e. from the drift direction) leaves an ionisation trail spanning the *full* gap in z and $g \tan \theta$ in x . (In this detector, as Sec. 5 shows, only the fraction of the column within $z_{\text{vis}} \approx 19.4$ mm of the mesh survives the drift with enough charge to be recorded — everywhere below, “the gap” as seen by the strips means this *visible column* z_{vis} , not the 30 mm electrode spacing.) Electrons drift to the mesh at velocity v_d , so the strip at horizontal distance Δx from the track’s mesh-crossing point receives its charge later by

$$\Delta t = \frac{\Delta z}{v_d} = \frac{\Delta x}{v_d \tan \theta}. \quad (1)$$

Fitting strip signal time versus strip position (amplitude-weighted straight line, anchored at the earliest hit) yields a slope whose inverse is

$$\frac{dx}{dt} = v_d \tan \theta \quad \implies \quad \tan \theta_{\text{det}} = \frac{1}{v_d} \frac{dx}{dt}. \quad (2)$$

Equation (2) is the micro-TPC angle estimator: it needs the drift velocity v_d as an external input, and conversely, once an external reference (θ_{ref} from M3) is available, the *correlation between* θ_{det}

and θ_{ref} measures v_d . That is the core idea exploited throughout this note. In the analysis code the raw fit slope is $s \equiv dx/dt$ in $\mu\text{m}/\text{ns}$; the detector-frame slope vector (s_x, s_y) is rotated by the alignment angle ($\theta_{\text{align}} \approx 90^\circ$) into the M3 frame before any comparison, so det-X pairs with ref-Y and vice versa.

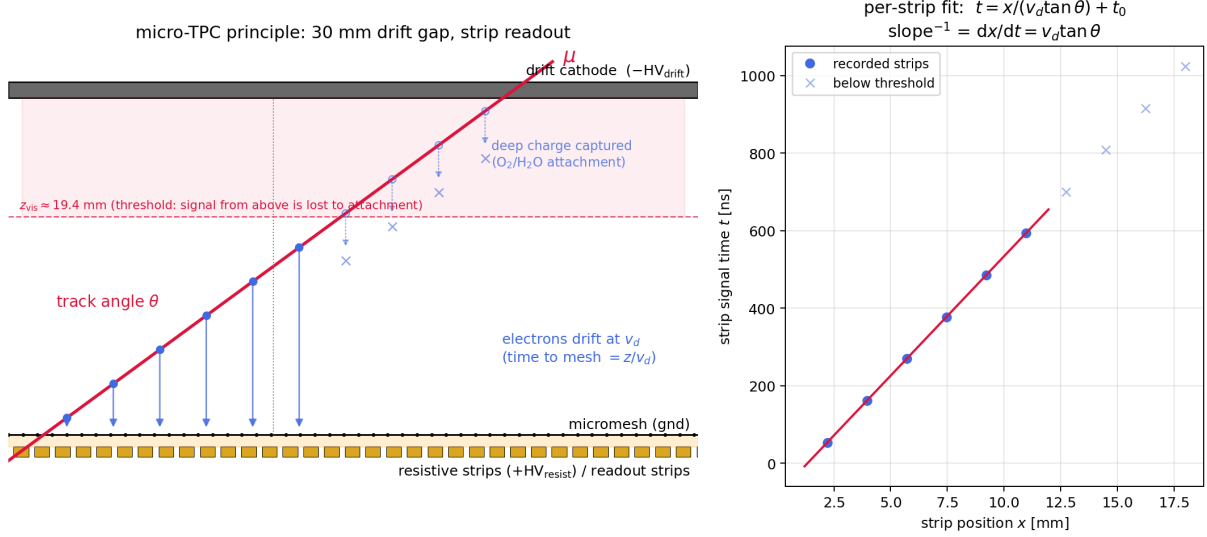


Figure 2: Micro-TPC principle. Left: an inclined muon (red) ionises the gas along the full 30 mm drift gap; electrons (blue) drift down at v_d and arrive at the mesh with delays proportional to their production height. Charge from above $z_{\text{vis}} \approx 19.4\text{ mm}$ (pink band) is lost to attachment during the drift (Sec. 5). Right: the per-event fit of strip time vs. strip position over the recorded strips; the inverse slope is $v_d \tan \theta$.

2.2 Why the micro-TPC mode matters here

Position resolution (Fig. 1) uses only the charge centroid and is insensitive to everything discussed below. The micro-TPC *slope*, however, is the detector’s unique selling point — per-plane track angles from a single thin detector — and it is exactly this quantity that shows the systematic effects analysed next.

3 Why the angle correlation is always off-diagonal

3.1 The observation

Figure 3 shows the production angle-correlation plot for the Saturday long run: reference angle θ_{ref} (M3) versus detector angle θ_{det} (Eq. 2, with v_d set by the production diagonal-projection scan). The correlation is strong — the micro-TPC works — but three features stand out, and they appear *identically in every detector of the June campaign* (det2 shown for comparison in Fig. 4):

1. the dense ridge runs *parallel* to the $y = x$ diagonal but is displaced *outward*: $|\theta_{\text{det}}| > |\theta_{\text{ref}}|$ by $\sim 5\text{--}6^\circ$;
2. there are essentially *no* events with $|\theta_{\text{det}}| \lesssim 5^\circ$ — an exclusion gap around the vertical;
3. events with $\theta_{\text{ref}} \approx 0$ form *horizontal arms* extending to large $|\theta_{\text{det}}|$ of either sign.

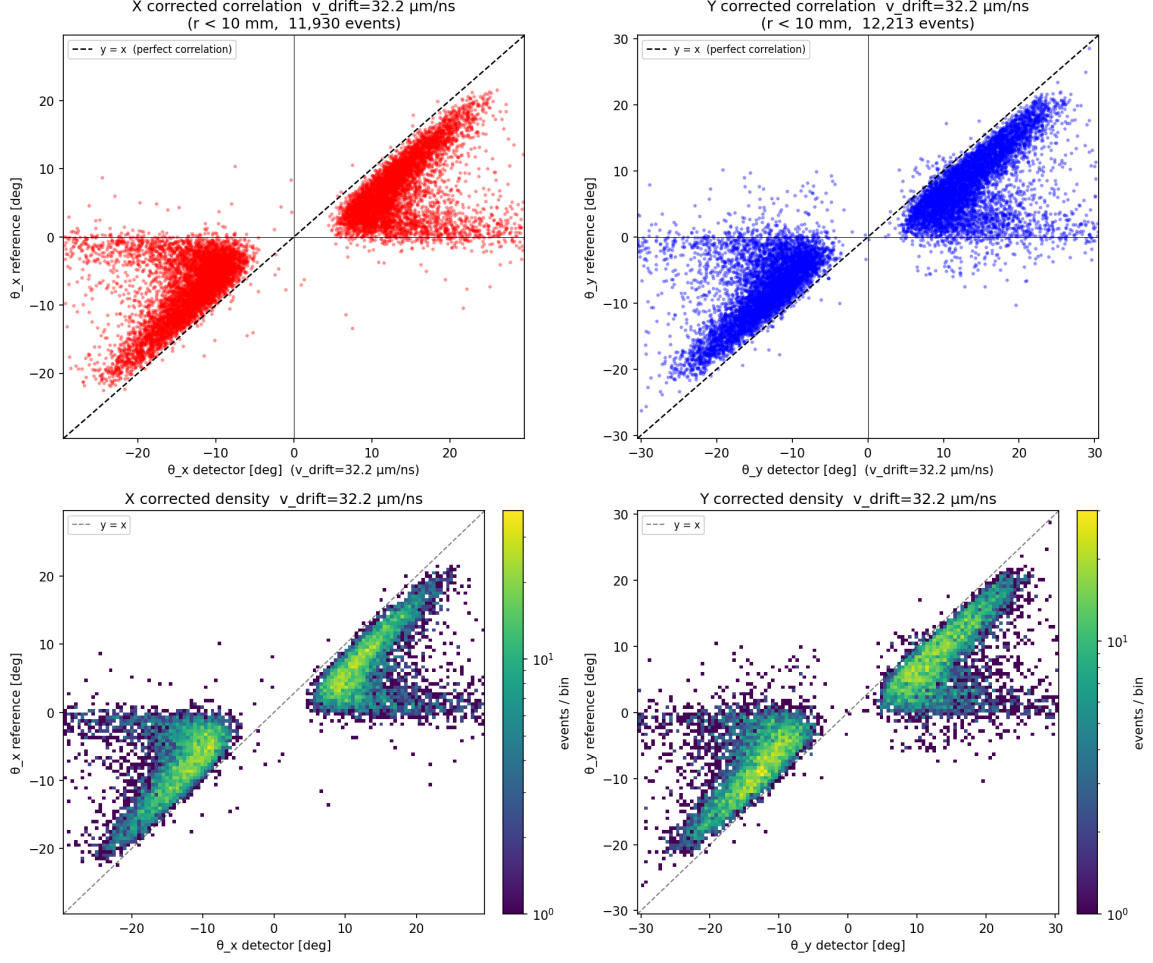


Figure 3: Saturday long run, production angle correlation (top: scatter; bottom: log-density) for the X (left) and Y (right) projections, using the v_d from the production σ -scan. The ridge is parallel to, but displaced outward from, the dashed $y = x$ diagonal; note the empty band $|\theta_{\text{det}}| \lesssim 5^\circ$ and the horizontal arms at $\theta_{\text{ref}} \approx 0$.

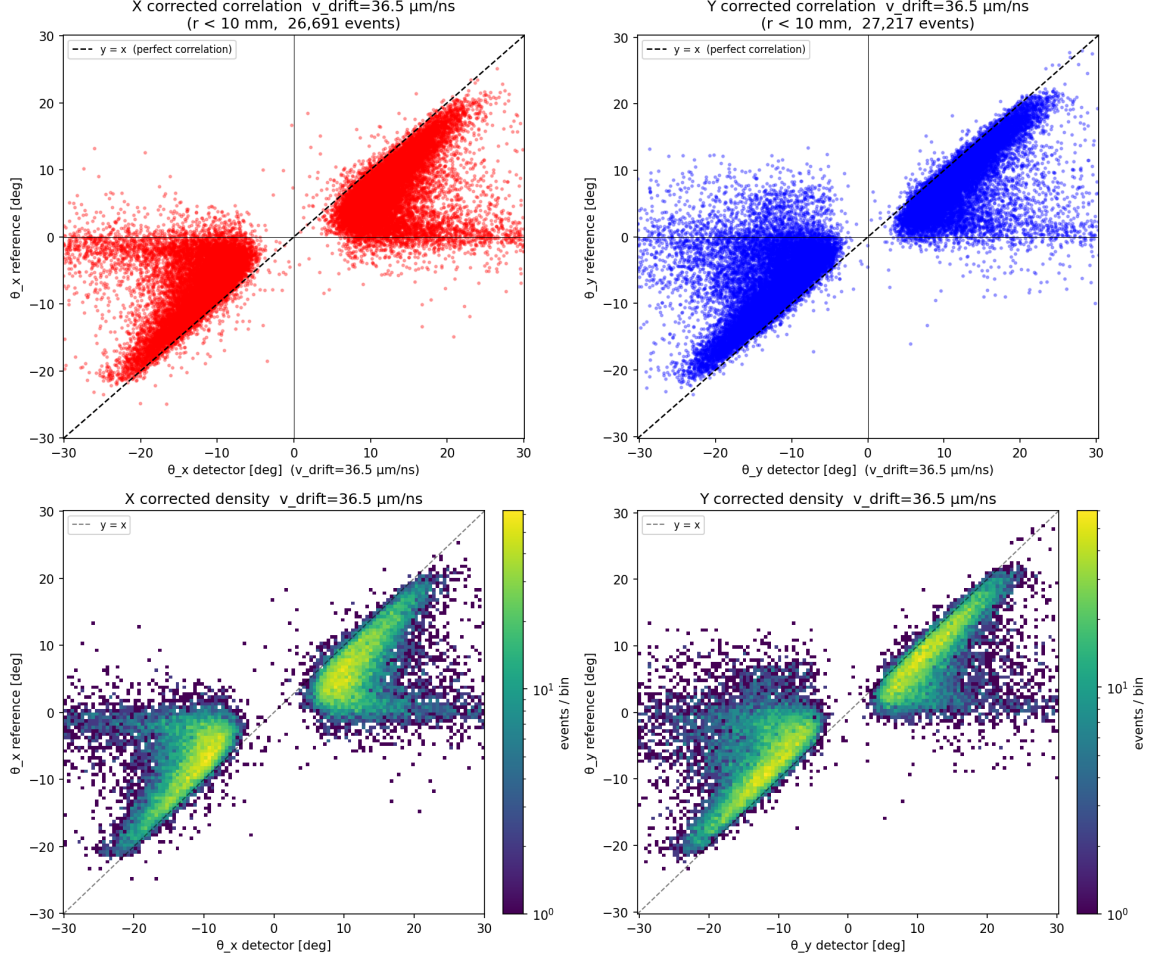


Figure 4: Same plot for *det2* (6-22 overnight run). The off-diagonal structure is identical — this is a property of the detector family, not of one chamber.

3.2 Why it is *not* a rotation or misalignment

A residual in-plane rotation is absorbed by the alignment (the rotation scan converges to 0.05° steps and the slope vector is rotated accordingly). More fundamentally, *any* rotation or constant angular offset would shift the correlation in *one* direction — up or down — whereas the observed displacement **flips sign with the track direction**: tracks leaning one way reconstruct too steep in that direction, tracks leaning the other way too steep in the other. The offset is odd in θ , i.e. it behaves like $\text{sign}(\theta) \cdot \text{const}$, which no rigid-body transformation produces. An out-of-plane tilt of the detector would also be a one-directional shift, and is excluded for the same reason.

3.3 The mechanism: fixed time span, inflated cluster width

The key geometric fact (Fig. 5): **every cosmic crosses the full drift gap**, so the recorded cluster always spans the entire *visible* column depth z_{vis} in z (the fixed depth at which drifting charge drops below threshold; Sec. 5). Consequently the *time* extent of every cluster is the same,

$$\Delta t_{\text{cluster}} \approx T_{\text{sat}} = z_{\text{vis}}/v_d, \quad \text{independent of } \theta, \quad (3)$$

while the *spatial* extent is

$$\Delta x_{\text{cluster}} \approx \underbrace{z_{\text{vis}} \tan \theta}_{\text{geometric}} + \underbrace{w}_{\text{non-geometric}}, \quad (4)$$

where w collects everything that widens the charge footprint on the strips without carrying drift-time information: charge spreading across the resistive layer, transverse diffusion, capacitive coupling to neighbours, and δ -electrons. The fitted slope is then, schematically (writing $z_v \equiv z_{\text{vis}}$),

$$\frac{dx}{dt} \approx \frac{z_v \tan \theta + w \text{sign}(\theta)}{z_v/v_d} = v_d \left(\tan \theta \pm \frac{w}{z_v} \right) \implies \boxed{\tan \theta_{\text{det}} \approx \tan \theta_{\text{ref}} \pm \frac{w}{z_{\text{vis}}}} \quad (5)$$

an **additive, angle-independent excess in tan-space** whose sign follows the track sign. (The sign attaches to w because the fit interprets the extra width as extra track length in the direction the track already leans; for a perfectly vertical track the sign is random — hence the horizontal arms.)

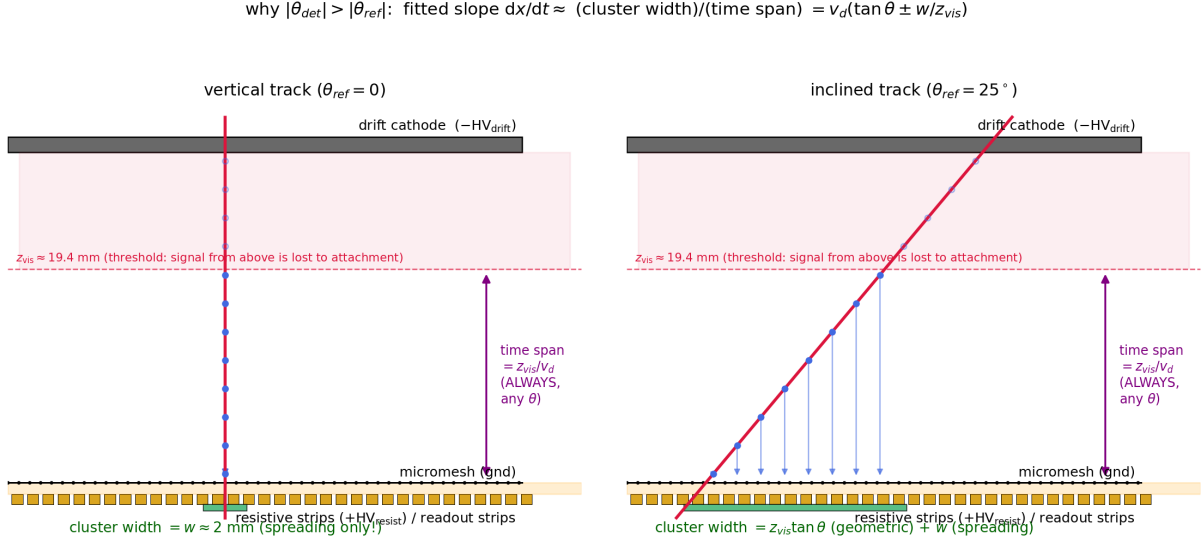


Figure 5: The off-diagonal mechanism. Left: a vertical track — the time span still covers the full visible column (z_{vis}/v_d), but the cluster width is pure spreading, $w \approx 2 \text{ mm}$; the fit converts w into a fake angle $\arctan(w/z_{\text{vis}}) \approx 6.5^\circ$ of random sign. Right: an inclined track — the width gains the geometric part $z_{\text{vis}} \tan \theta$, but w still adds on top, always *increasing* $|\theta_{\text{det}}|$.

This single mechanism explains all three observations at once:

- **parallel outward ridge:** Eq. (5) is a constant displacement w/z_{vis} in tan-space; converted to degrees it is a nearly constant $\sim \arctan(w/z_{\text{vis}})$ shift over the populated range;
- **exclusion gap:** no cluster can fit a slope smaller than $\sim w/z_{\text{vis}}$, because even a vertical track's cluster is w wide over a full-column time span — $|\theta_{\text{det}}| \gtrsim \arctan(w/z_{\text{vis}})$ always;
- **horizontal arms:** for $\theta_{\text{ref}} \approx 0$ the geometric width vanishes, the sign of the fitted slope is set by noise, and the magnitude is again $\sim w/z_{\text{vis}}$ or larger (the events that pass the ≥ 4 -strip cut at small angle are precisely those with the largest spreading).

And because w (a resistive-layer/diffusion property) and z_{vis} (set by the shared gas supply and thresholds, Sec. 5) are common to the whole detector family, **every chamber shows the same off-diagonal pattern** — exactly what is observed.

3.4 Quantitative verification

All of this is testable in the data of the 2.4 h run (`13_tpc_angle_bias.py`); Figs. 6 and 7:

- **the excess is flat in tan-space:** the median of $|\tan \theta_{\text{det}}| - |\tan \theta_{\text{ref}}|$ is constant at $w/z_{\text{vis}} = 0.115$ for $|\theta_{\text{ref}}| \gtrsim 6^\circ$, in both planes (Fig. 6, bottom row). $\arctan(0.115) = 6.6^\circ$ — the observed ridge displacement;

- **wider clusters carry more bias:** splitting by cluster size (4–6 / 7–9 / ≥ 10 strips) orders the excess exactly as the spreading picture predicts;
- **the time span saturates:** the median cluster time span rises to a plateau $T_{\text{sat}} \approx 690$ ns for $|\theta_{\text{ref}}| > 10^\circ$ — the full-gap drift time (Fig. 7, top row). (Its decrease towards $\theta = 0$ is a per-strip integration effect: a vertical track’s charge lands on few strips whose fitted times average over the column, shrinking the strip-to-strip span.);
- **the spatial width has a floor:** the median cluster extent at $\theta_{\text{ref}} \approx 0$ is 3–5 mm where geometry predicts ≈ 0 ; at larger angles it grows parallel to the geometric expectation $z_{\text{vis}} \tan \theta$, offset by the same floor (Fig. 7, bottom row).

The ridge-fit intercepts (Sec. 4) give the same number from a completely different projection of the data: $|b|/v = w/z_{\text{vis}} = 0.115$, i.e. $w = 2.2$ mm (≈ 3 strips) for $z_{\text{vis}} = 19.4$ mm. Finally, overlaying the parameter-free prediction of Eq. (5) on the measured density (Fig. 8) tracks the ridge in both planes.

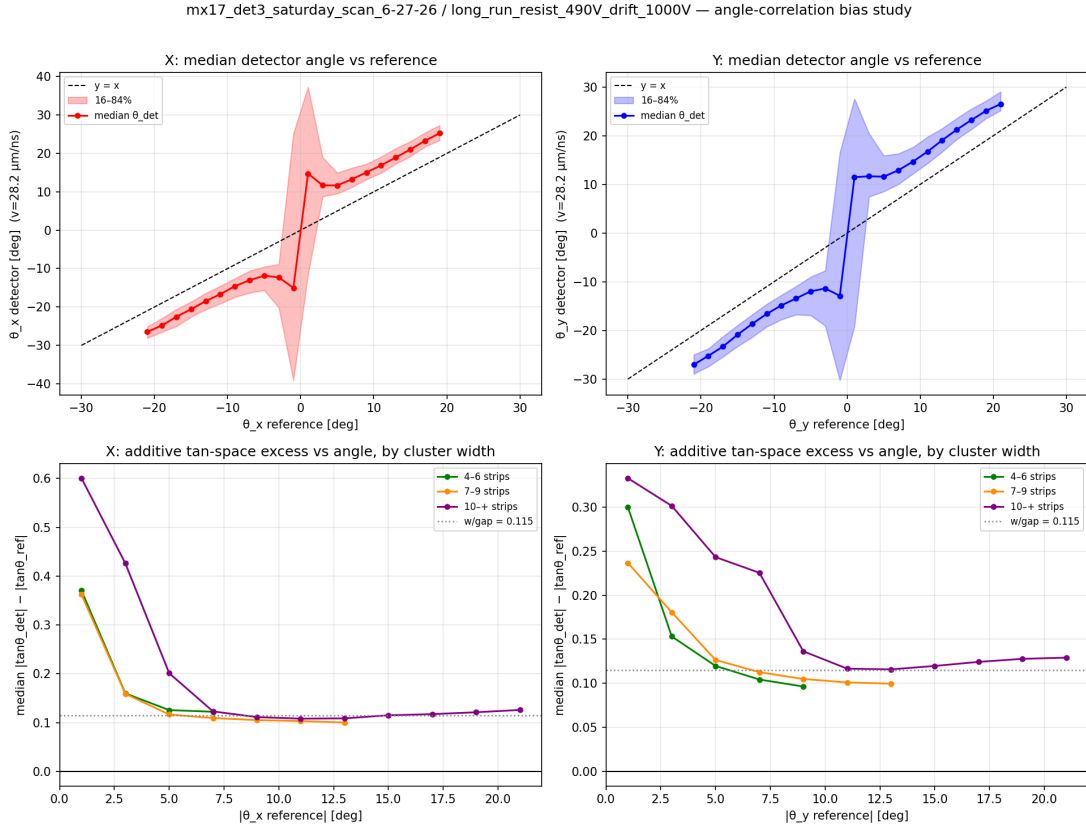


Figure 6: Bias study on the 2.4h run. Top: median θ_{det} vs θ_{ref} — parallel to the diagonal, displaced outward, with the sign flip through zero. Bottom: the additive tan-space excess $|\tan \theta_{\text{det}}| - |\tan \theta_{\text{ref}}|$ vs $|\theta_{\text{ref}}|$, split by cluster size; it is flat at $w/z_{\text{vis}} = 0.115$ (dotted) above $\sim 6^\circ$ and grows for wide clusters. The rise at small angles is the ≥ 4 -strip selection (only heavily-spread clusters survive).

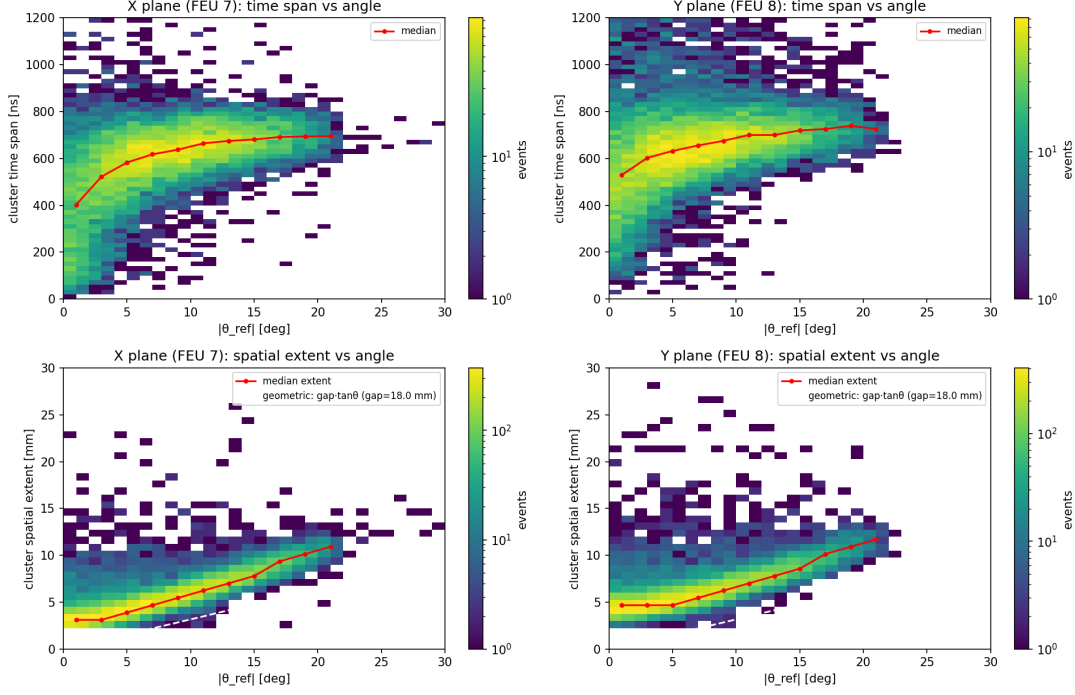


Figure 7: Cluster geometry vs track angle (2.4 h run). Top: cluster *time* span — saturates at $T_{\text{sat}} \approx 690$ ns (the drift time across the visible column) for inclined tracks. Bottom: cluster *spatial* extent — a 3–5 mm non-geometric floor at $\theta = 0$, then growth parallel to the geometric $z_{\text{vis}} \tan \theta$ (white dashed).

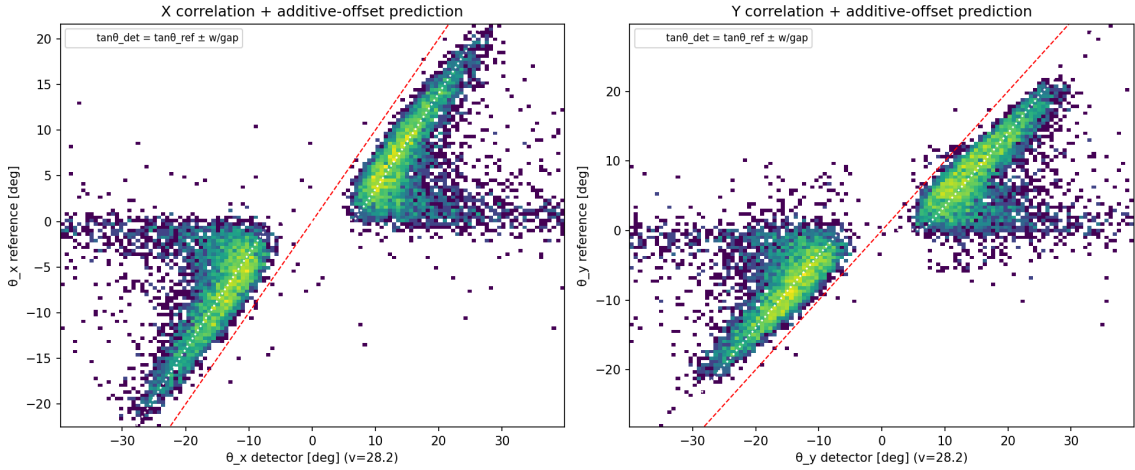


Figure 8: Measured correlation density with the parameter-free prediction of Eq. (5), $\tan \theta_{\text{det}} = \tan \theta_{\text{ref}} \pm w/z_{\text{vis}}$ (white dotted), using the independently fitted $w/z_{\text{vis}} = 0.115$. The red dashed line is $y = x$.

3.5 Consequences

1. **The production σ -scan drift velocity is biased.** The production estimator (Fig. 9) scans v_d and minimises the width of $|\theta_{\text{ref}}| - |\theta_{\text{det}}|$; the additive offset pulls its minimum high ($v_d = 30.5/34.0 \mu\text{m/ns}$ for X/Y on this run versus the unbiased $28.1 \mu\text{m/ns}$ of Sec. 4). Its $|\cdot|$ -folding also mixes the sign-following offset into the width. Use the ridge fit instead.
2. **A first-order angle correction exists:** subtract $\text{sign}(\theta_{\text{det}}) w/z_{\text{vis}}$ in tan-space. It sharp-

ens inclined tracks but cannot recover $|\theta| \lesssim 5^\circ$, where the angular information is genuinely destroyed by spreading.

3. **Angular resolution** (after slicing in θ_{ref} , i.e. with the local bias absorbed): $\sigma_\theta \approx 1.5 - 2^\circ$ per plane for $|\theta| \gtrsim 5^\circ$, degrading to $\sim 6^\circ$ near normal incidence (Fig. 10).

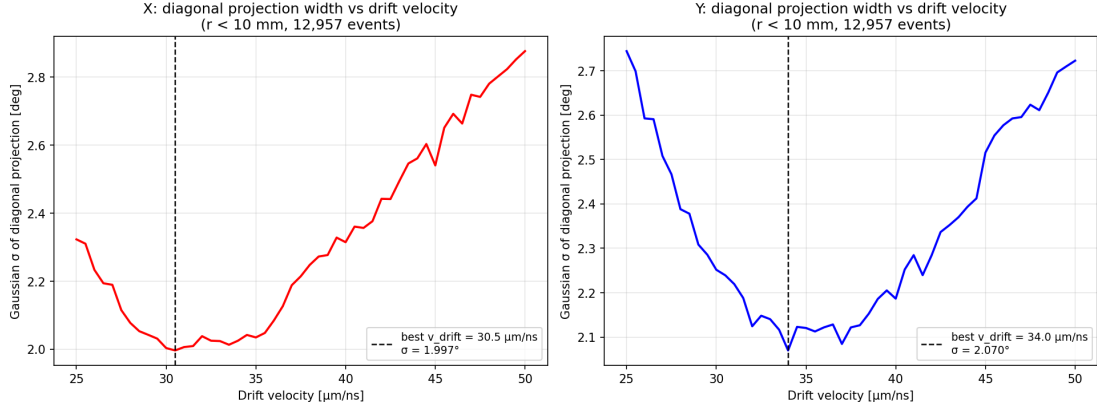


Figure 9: Production diagonal-projection σ -scan on the Saturday run. The minima (X: 30.5, Y: 34.0 $\mu\text{m/ns}$) are biased high by the additive offset; kept here for reference only.

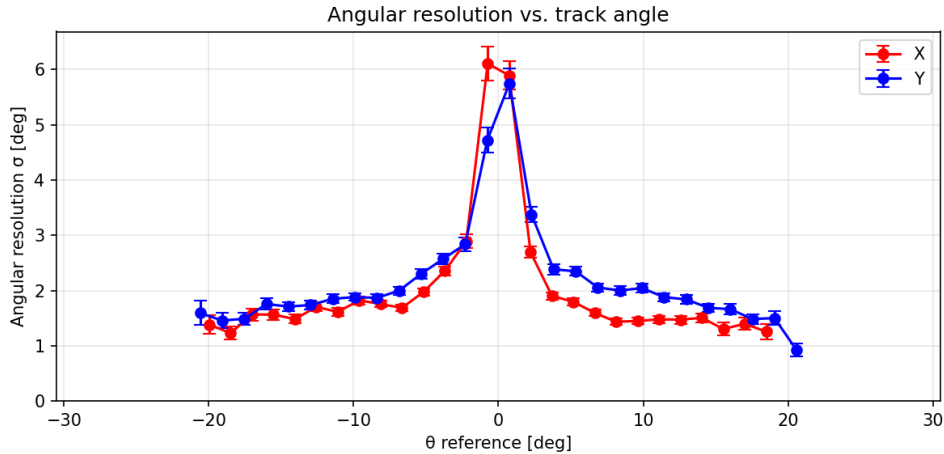


Figure 10: Micro-TPC angular resolution vs track angle (2.4 h run): $\sigma_\theta \approx 1.5 - 2^\circ$ per plane for inclined tracks; the method collapses below $\sim 5^\circ$ as predicted by the spreading mechanism.

4 Drift-velocity measurement from the drift scan

4.1 A bias-free estimator: the ridge fit

Rewriting Eq. (5) in terms of the *raw fitted slope* $S \equiv v_d \tan \theta_{\text{det}}$ (in $\mu\text{m/ns}$, after rotation to the M3 frame):

$$S = v_d \tan \theta_{\text{ref}} \pm v_d \frac{w}{z_{\text{vis}}}. \quad (6)$$

This is a straight line in the $(\tan \theta_{\text{ref}}, S)$ plane whose **slope is the drift velocity** and whose intercept absorbs the entire spreading offset. A per-track-sign straight-line fit (iteratively 3σ -clipped, fit window $0.06 < |\tan \theta_{\text{ref}}| < 0.55$) therefore measures v_d *without* the bias that afflicts the σ -scan — no assumption about w , no folding of signs, and no drift-velocity prior. Four independent fits (X/Y plane $\times$ track sign) are combined by weighted mean; their spread is the

quoted fit systematic. The M3 angular resolution ($\sim 0.05^\circ$) is negligible, so there is no regression dilution.

Two robustness checks: the result is insensitive to the minimum cluster size (4/5/6 strips give identical v_d), and moving the fit window shifts v_d by up to $\pm 10\%$ (the ridge has a slight curvature — the additive model is not perfect at that level). The default window is anchored by the independent time-span cross-check below agreeing to 1%; we assign $\sim 5\%$ to the absolute scale, while the *shape* of $v_d(\text{HV})$, taken with a fixed window, is much tighter.

4.2 Cross-check: the visible-column drift time

The time-span plateau T_{sat} (Fig. 7) is an independent $1/v_d$ observable: $T_{\text{sat}} = z_{\text{vis}}/v_d$ with the *same* z_{vis} at every HV (the visible column is a fixed depth, Sec. 5). It shares no systematics with the ridge fit except the time calibration, but note it is not fully independent as a *velocity* scale: z_{vis} itself is calibrated from $v_d \cdot T_{\text{sat}}$ of the long run. What it does test, powerfully, is that $v_d(\text{HV}) \cdot T_{\text{sat}}(\text{HV})$ is one constant across the scan. Table 2 and Fig. 11 show both quantities.

4.3 Results

drift HV [V]	v_d ridge [$\mu\text{m}/\text{ns}$]	T_{sat} [ns]	$z_{\text{vis}}/T_{\text{sat}}$ [$\mu\text{m}/\text{ns}$]	note
100	—	—	—	invalid: drift time $\gg$ readout window
300	5.70 ± 0.54	(1198)	—	window truncation marginal
500	12.77 ± 0.43	(1361)	—	T_{sat} truncated; ridge OK
700	19.45 ± 0.53	992	19.6	
900	26.17 ± 0.84	754	25.7	
1000	28.12 ± 0.65	690	28.1	2.4 h long run
1100	29.44 ± 0.47	661	29.4	

Table 2: Drift-velocity measurements (14.drift_velocity_scan.py). Errors combine statistics and the 4-fit spread. $z_{\text{vis}}/T_{\text{sat}}$ uses the fixed $z_{\text{vis}} = 19.4\text{ mm}$ determined from the long run; its agreement with the ridge fit at every valid point is the statement that $v_d \cdot T_{\text{sat}}$ is the same constant at all fields.

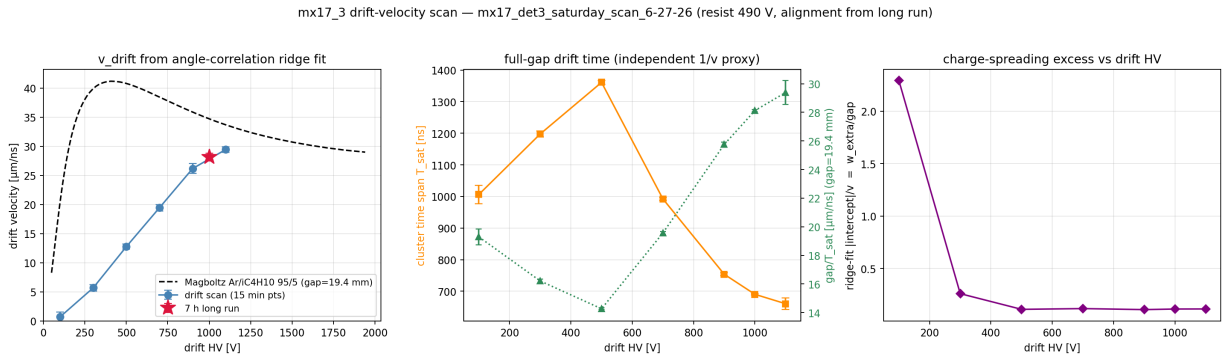


Figure 11: Left: v_d vs drift HV from the ridge fit; the 2.4 h run (star) falls exactly on the 15-minute scan curve, and the dashed line is the Magboltz prediction for *clean* gas (see Sec. 6). Middle: the time-span plateau T_{sat} (orange) and the derived $z_{\text{vis}}/T_{\text{sat}}$ (green) — at 300–500 V the span rails against the readout window instead of following $1/v_d$. Right: the ridge-fit intercept w/z_{vis} — constant at 0.115 for $\geq 500\text{ V}$: the spreading is a detector property, independent of drift field.

Key points:

- **Statistics are not a limitation.** You were right that the correlation measures v_d — and even the 15-minute points give 1–3 % statistical errors, because the ridge fit uses every inclined track, not just the width of a projection.
- **Internal consistency** is excellent: $v_d \cdot T_{\text{sat}}$ is one constant wherever the span is not window-limited; the 2.4 h point sits on the scan curve; w/z_{vis} is field-independent.
- **The low-HV points die by window truncation**, not detector effects: at 100 V the full-gap drift time ($\gtrsim 10 \mu\text{s}$ for 30 mm) vastly exceeds the $1.92 \mu\text{s}$ DREAM window — most charge simply arrives after the readout closes. The same truncation (compounded at low field by stronger attachment losses, Sec. 5) explains the efficiency turn-on of the drift scan (Fig. 12): 44 % at 100 V rising to a 83 % plateau at ≥ 700 V. This is the quantitative version of the June observation “det3 needs drift ≥ 700 V”.

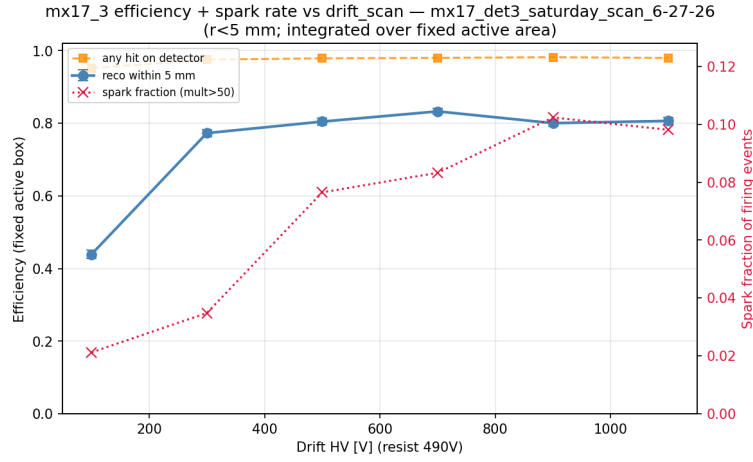


Figure 12: Reconstruction efficiency vs drift HV (drift scan, fixed active box). The turn-on tracks the fraction of the drift column whose charge arrives inside the readout window (and survives attachment); the plateau begins once the surviving column’s drift time fits in the window.

5 The drift gap: 30 mm mechanical, 19.4 mm visible

The middle panel of Fig. 11 shows the quantity $v_d \cdot T_{\text{sat}}$: it is one constant, 19.4(2) mm, across a $1.6\times$ range of drift field. A first reading of that number took it to be the drift gap itself — 10.6 mm short of the 30 mm design. The mechanical gap, however, is known to be close to 30 mm (certainly > 27 mm), and a dedicated test (`17_gap_attachment_test.py`, Fig. 14) resolves the conflict cleanly: **the gap is 30 mm; 19.4 mm is the depth of drift column whose signal survives above threshold**. Electrons produced higher up *do* arrive — with exponentially attenuated amplitude — because they are progressively captured (attachment) in the contaminated gas of Sec. 6.

5.1 What $v_d \cdot T_{\text{sat}}$ actually measures

T_{sat} is the plateau of the *cluster* time span, and a cluster only contains strips above threshold. The product $v_d \cdot T_{\text{sat}}$ is therefore the depth of the *visible* column, z_{vis} : the height at which the per-strip amplitude falls below the hit threshold. Its constancy with drift field says that this threshold contour sits at a fixed *depth* — which is exactly what a depth-dependent survival probability (attachment over a fixed length scale) produces. Two properties of the product remain valuable: it is immune to time-calibration errors (velocity $\propto 1/k$, span $\propto k$ under any time rescaling k), and it is the correct normalisation for the micro-TPC geometry of Secs. 3–4, which live entirely inside the visible column.

5.2 The discriminating test

If the column geometrically ended at 19.4 mm there would be *nothing but noise floor* beyond T_{sat} in the strip arrival-time distribution. If instead the gap is 30 mm with progressive charge loss, the distribution must carry a suppressed tail that terminates at the 30 mm drift time, and the strip *amplitude* must decay with drift depth. Figure 14 tests this per drift-scan point, with all times converted to depth via $z = v_d t$:

- **the tails do not end at 19.4 mm (top left):** every HV point shows the same shape — flat plateau to 19.4 mm, then an exponential shoulder falling ~ 2 orders of magnitude and terminating at 28–29 mm where the readout window permits (900–1100 V; the 700 V endpoint at 25.7 mm and the 500 V one at 20 mm are window-truncated lower bounds). The tail endpoint sits at a fixed *depth*, i.e. it scales as $1/v_d$ in time (bottom row) — a geometric edge at ≈ 30 mm, softened by the threshold;
- **the amplitude decays with depth, identically at every field** (Fig. 13 and top right of Fig. 14): versus drift *time* the decay timescale changes with HV, but versus drift *distance* the 700–1100 V curves collapse onto one exponential — $\lambda \approx 10$ mm fitted over 8–24 mm, 13–15 mm inside the visible column — declining by an order of magnitude between mid-column and 30 mm. Signal loss is a function of *how far the charge drifts*, not of time or field — the fingerprint of capture on electronegative impurities (O_2), with transverse diffusion contributing a milder per-strip dilution on top;
- **the “cliff” is a threshold crossing:** the arrival-time distribution stays flat while the amplitude distribution clears the hit threshold, then falls over a few bins as the median approaches it — steeper than the amplitude decay itself because the occupancy integrates the amplitude distribution above a cut. The earlier reading of this edge as a hard geometric boundary (Fig. 15) missed that the exponential tail beyond it ends precisely at the 30 mm drift time (dotted line in the right panel).

mx17_3 — attachment signature: amplitude decay is a function of drift distance, not time (mx17_det3_saturday_scan_6-27-26, inclined tracks)

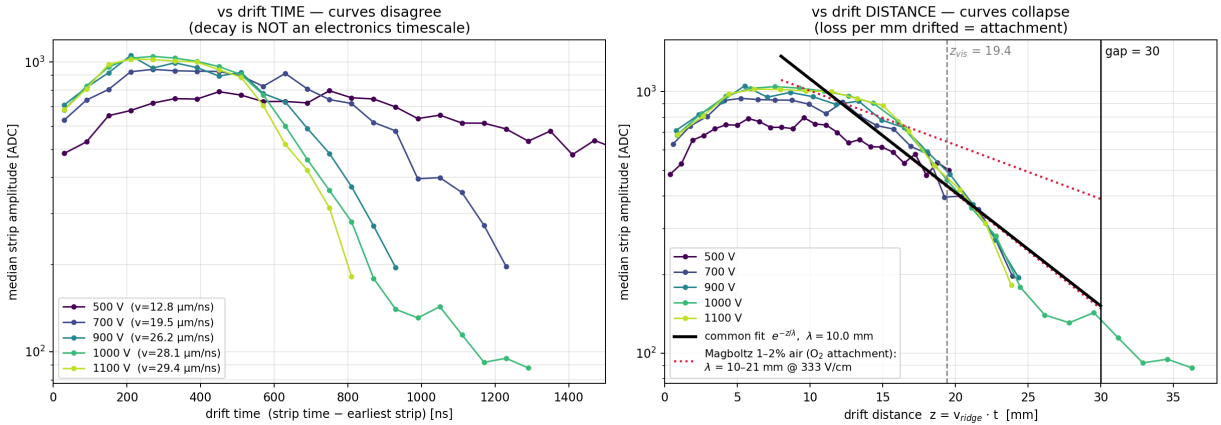


Figure 13: The attachment signature in one plot (19_amplitude_attachment_plot.py). Median strip amplitude for inclined tracks, per drift HV. Left: versus drift *time* — the decay timescale changes with drift field (1100 V dies by 800 ns while 500 V is still flat at 1400 ns), so it is not an electronics timescale (ion tail, shaping, undershoot would be identical curves). Right: the same data versus drift *distance* $z = v_d t$ — the 700–1100 V curves collapse onto a single exponential, $\lambda \approx 10$ mm over 8–24 mm (13–15 mm when fitted inside the visible column only), matching the Magboltz O_2 -attachment band for 1–2 % air (red dotted). Loss per millimetre drifted is electron capture.

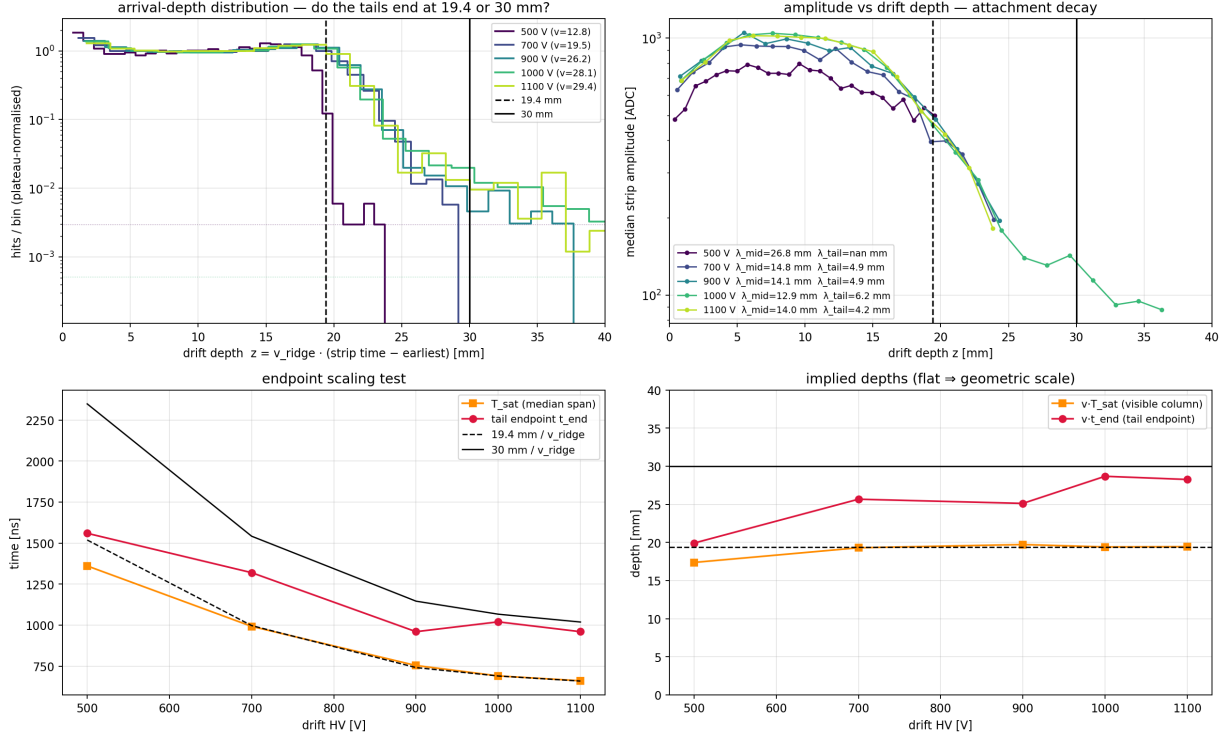


Figure 14: Gap discrimination (`17_gap_attachment_test.py`). Top left: strip arrival-*depth* distributions ($z = v_d t$, plateau-normalised, log scale) per drift HV; all curves share the plateau end at 19.4 mm (dashed) *and* a suppressed tail reaching ≈ 30 mm (solid) where the readout window allows. Top right: median strip amplitude vs depth — one universal decay curve for all fields. Bottom: the time-span plateau T_{sat} and the tail endpoint versus HV, in time (left) and depth (right); the endpoint tracks the 30 mm/ v_d expectation while T_{sat} tracks 19.4 mm/ v_d .

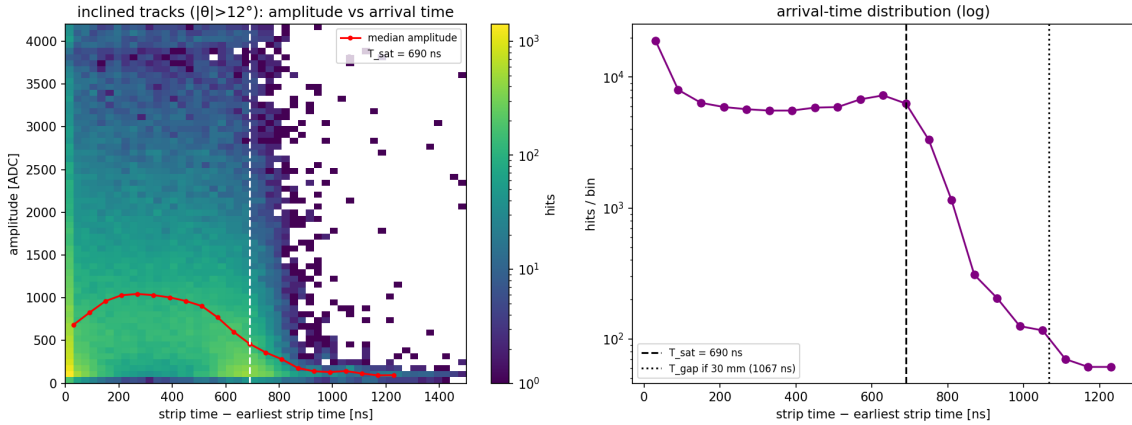


Figure 15: Inclined tracks, 2.4h run (the plot that, on closer reading, decides the question). Left: strip amplitude vs arrival time; the median (red) decays steadily with drift time. Right: the arrival-time distribution — flat to $T_{sat} = 690$ ns (dashed), then an *exponential* fall that terminates at the 30 mm drift time of 1067 ns (dotted) before flattening into the noise floor.

5.3 Consistency of the other observables

The single-strip pulse widths (Fig. 16) measure the duration over which a vertical track's lead strip stays above threshold — i.e. the *surviving* column's drift time. Their ~ 700 ns scale is consistent

with the attachment picture and carries no independent information about the mechanical gap; it is no longer evidence for a 19.4 mm electrode spacing.

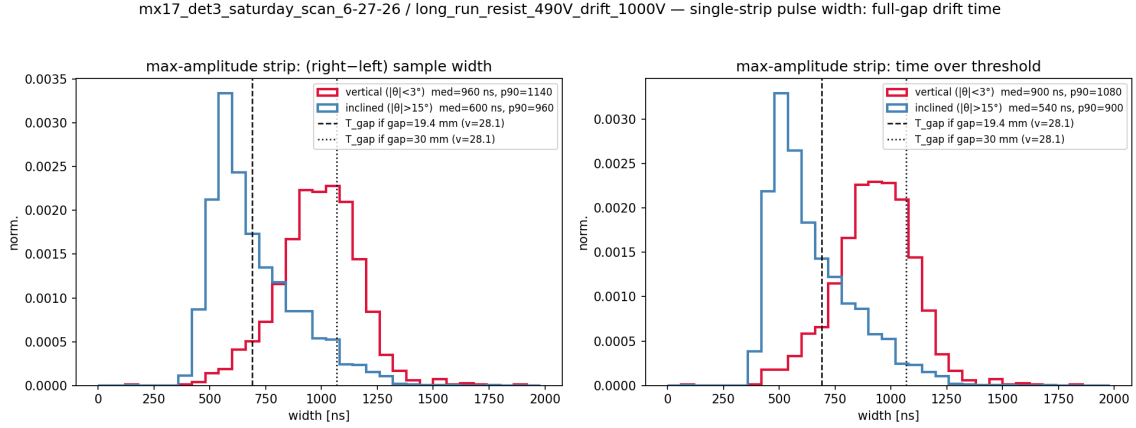


Figure 16: Single-strip pulse width for vertical ($|\theta| < 3^\circ$, red) versus inclined ($|\theta| > 15^\circ$, blue) tracks; the vertical-track widths reflect the drift time of the *visible* column (≈ 700 ns at 1000 V), as the attachment picture predicts.

5.4 Interpretation and consequences

- **Field scale:** with $g = 30$ mm the drift field is $E = V_{\text{drift}}/3.0$ cm (1000 V $\rightarrow$ 333 V/cm); Sec. 6 uses this throughout. The ridge-fit velocities are unaffected — the fitted slope is a local dx/dt inside the visible column.
- z_{vis} **is a gas-quality number, not a detector constant:** it is where the survival probability $\exp(-z/\lambda_{\text{att}})$ meets the threshold. Cleaning the gas should push the visible column towards the full 30 mm — lengthening T_{sat} (mind the 1.92 μ s window: 30 mm at 28 μ m/ns is 1067 ns, still comfortable), increasing the collected charge, and shrinking the off-diagonal offset w/z_{vis} of Sec. 3.
- **Charge budget:** roughly a third of each track’s ionisation is currently lost in transit, and what survives from mid-column is attenuated — a direct efficiency and signal-to-noise cost on top of the drift-speed problem.
- The 60 ns DREAM sampling period assumption still deserves a one-line confirmation; it rescales v_d and all times together but no conclusion here depends on it.

6 Gas diagnosis: air/moisture contamination of the Ar/iso 95/5

6.1 The discrepancy

With the 30 mm gap of Sec. 5 the field scale is $E = V_{\text{drift}}/3.0$ cm (1000 V $\rightarrow$ 333 V/cm), and the measured $v_d(E)$ can be confronted with Magboltz (Garfield++ **MediumMagboltz**, Saclay pressure 745.8 Torr, 293 K). For *clean* Ar/iC₄H₁₀ 95/5 the prediction peaks at 41 μ m/ns near 210 V/cm and *decreases* slowly above; the measurement instead rises monotonically over the whole scan, is a factor ~ 3 low at 167 V/cm (12.8 vs ~ 40 μ m/ns), and reaches only 28 μ m/ns at the operating field where clean gas predicts 39 μ m/ns. The shape disagreement is the crucial part: no rescaling of times, fields or gap can turn a curve that peaks at 210 V/cm into one still rising at 370 V/cm. This is a *gas property*.

6.2 Candidate hypotheses: velocity alone is ambiguous

Magboltz was run for a set of candidate gases over the drift-field range (scripts `mm_drift_velocity_candidates.py`, `mm_attachment_{table,air_candidates}.py` in `garfield_sim/`); Fig. 17 and Table 3:

hypothesis	RMS dev. [$\mu\text{m}/\text{ns}$]	verdict
Ar/CO ₂ 90/10 (wrong bottle)	1.8	best $v_d(E)$ — <i>excluded</i> by gain & attachment
Ar/iso 95/5 + 1% H ₂ O	4.4	consistent (best fit $\approx 1\text{--}1.5\%$)
Ar/iso 95/5 + 2% H ₂ O	8.5	too much suppression
Ar/iso 95/5 + 2.5–3% H ₂ O	12–14	excluded
Ar/iso 95/5 + 1–2% dry air	17–19	no v_d suppression (N ₂ /O ₂ too dilute)
Ar/iso 95/5 (nominal, clean)	20.2	excluded
Ar/iso 90/10, 85/15, 80/20	12–20	iso-rich gas stays <i>fast</i> : excluded

Table 3: RMS deviation between each Magboltz candidate and the measured points, with the 30 mm field scale. (Under the superseded 19.4 mm scale the ranking had favoured 2–2.5% H₂O; the gap revision moves the best fit to $\approx 1\text{--}1.5\%$ H₂O.)

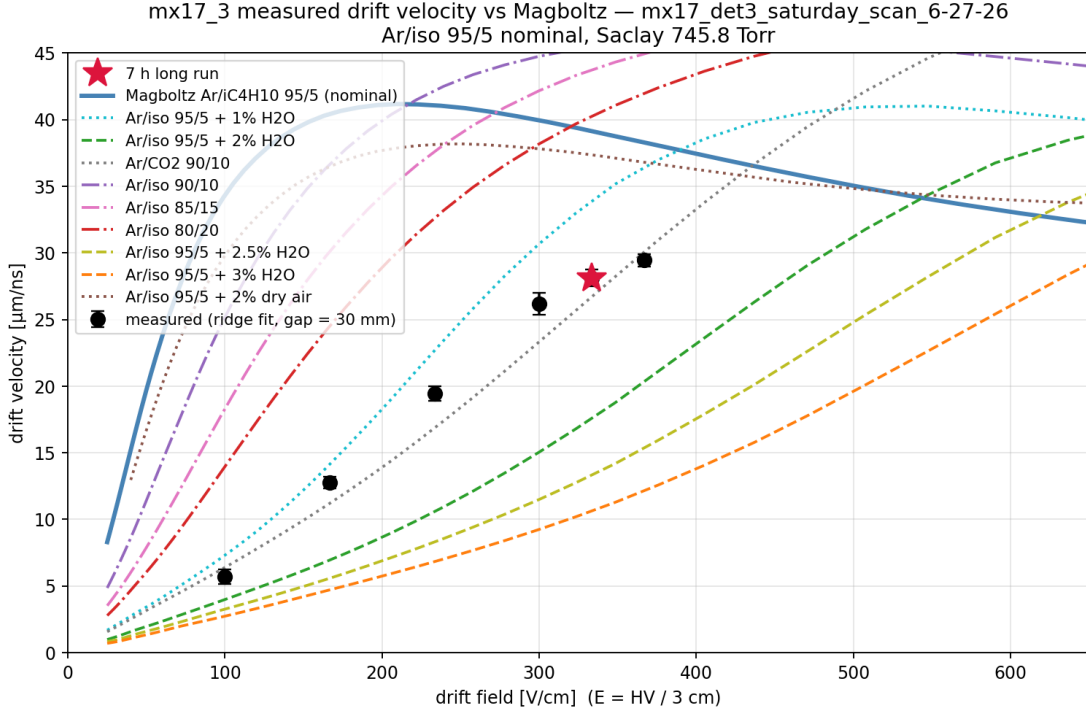


Figure 17: Measured $v_d(E)$ (black; star = 2.4 h run) against Magboltz predictions, $E = V/3.0$ cm. Clean Ar/iso 95/5 (blue) is excluded in shape and magnitude; iso-rich mixtures (dash-dotted) remain fast; dry air (brown dotted) barely changes v_d ; the data run between the 1–2% H₂O curves and, coincidentally, along Ar/CO₂ 90/10 (grey dotted) — the velocity alone cannot separate the last two.

The velocity curve leaves *two* viable candidates — a wrong bottle of Ar/CO₂ 90/10, or water contamination at the 1–1.5% level — and they are separated by the two other observables in hand:

Amplification. A Magboltz Townsend-coefficient comparison over the 150 μm amplification gap (`mm_townsend_compare.py`) puts every iso-gain point of Ar/CO₂ 90/10 about 50 V *above* Ar/iso 95/5 (gain 10^2 : 528 V vs 475 V; gain 10^3 : 631 V vs 587 V). The observed turn-on at 425 V

with full plateau by 455 V (Sec. 7) matches the Ar/iso expectation; with Ar/CO₂ the whole efficiency curve would sit ~ 50 V higher. Percent-level H₂O/air, by contrast, leaves the gain curve essentially untouched.

Attachment. Clean Ar/CO₂ does not attach electrons at drift fields ($\eta = 0$ in Magboltz) — it cannot produce the amplitude decay of Fig. 14. Neither, notably, can water alone: Magboltz gives $\eta = 0$ for all Ar/iso + H₂O mixtures at these fields (water’s dissociative-attachment threshold, ~ 5 eV, is far above drift energies). The measured decay needs **oxygen**.

6.3 The attachment cross-check closes on O₂

Figure 18 compares the measured amplitude-decay length from the drift scan with the Magboltz attachment length $1/\eta(E)$ for air-doped mixtures. The measured mid-column $\lambda \approx 13 - 15$ mm at 230–370 V/cm falls between the 1%-air (0.21 % O₂, $\lambda \approx 16 - 21$ mm) and 2%-air (0.42 % O₂, $\lambda \approx 10$ mm) predictions: **an O₂ fraction of ~ 0.2 – 0.4 % reproduces the observed signal loss**. (The measured λ also absorbs some per-strip dilution from transverse diffusion, so the true attachment length is slightly longer and the implied O₂ correspondingly at the lower end.)

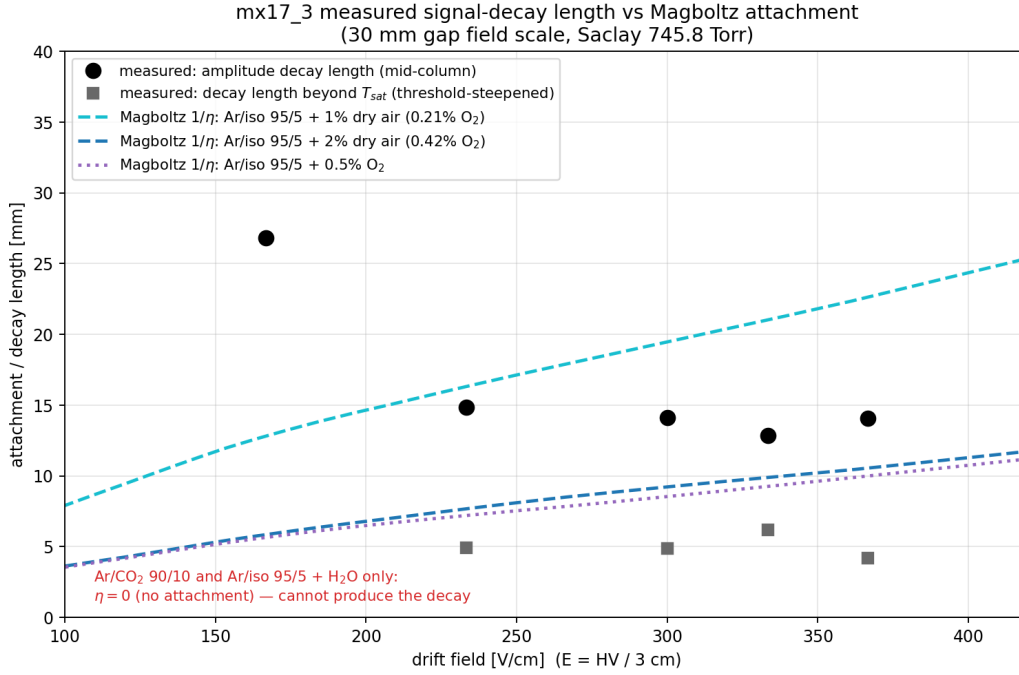


Figure 18: Measured signal-decay length vs drift field (black: mid-column; grey: beyond T_{sat} , steepened by the threshold), compared with Magboltz attachment lengths $1/\eta$ for O₂-doped Ar/iso 95/5. Mixtures without O₂ (clean Ar/CO₂, Ar/iso + H₂O only) predict no attachment at all.

6.4 Interpretation: one contamination, both anomalies

Everything points at a single cause — **air and moisture in the drift gas** at the percent level:

- ~ 1 – 1.5 % H₂O reproduces the slow, monotonically rising $v_d(E)$ (water’s huge dipole cross-section thermalises the electrons at low field). That is ~ 40 – 70 % relative humidity worth of water — ordinary damp-air/outgassing levels, not a flooded line;
- ~ 0.2 – 0.4 % O₂ (i.e. ~ 1 – 2 % air-equivalent) reproduces the attachment that truncates the visible drift column at 19.4 mm (Sec. 5);
- the amplification is insensitive to impurities at this level — consistent with the resist-HV scans matching clean-gas gain;

- water in excess of the O_2 -implied air fraction is expected: H_2O permeates plastics and out-gasses from glued structures far faster than O_2 enters, the classic signature of an unflushed or under-flushed detector volume.

Likely entry points: an unflushed detector volume, permeable or long tubing, very low flow rate, or a leaky fitting. Decisive follow-ups:

- **fresh-gas re-scan** of one detector after a long flush: at 1000 V the velocity should rise from 28 towards $\sim 39 \mu\text{m}/\text{ns}$, *and* the visible column should extend from 19.4 mm towards the full 30 mm (with the amplitude decay flattening) — a one-hour measurement with the ridge fit plus the Fig. 14 analysis;
- humidity *and* O_2 probes on the gas exhaust; a flow/leak check of the line.

Beyond drift speed, the contamination costs signal: a third of each track's charge is captured before reaching the mesh, and the enhanced transverse diffusion feeds the charge-spreading width w of Sec. 3. Cleaning the gas should improve efficiency, signal-to-noise, micro-TPC angle quality, and the small-angle exclusion gap simultaneously.

7 Resist-HV scans

Two resist scans bracket the operating point at drift 1000 V (Fig. 19): a wide scan (425–525 V, 11 points) and a fine repeat (460–520 V). Both use the long-run alignment with a translation refit per point; efficiency is counted in a fixed active box with the standard $R = 5$ mm matching, and the spark fraction (events with > 50 strips firing) is overlaid.

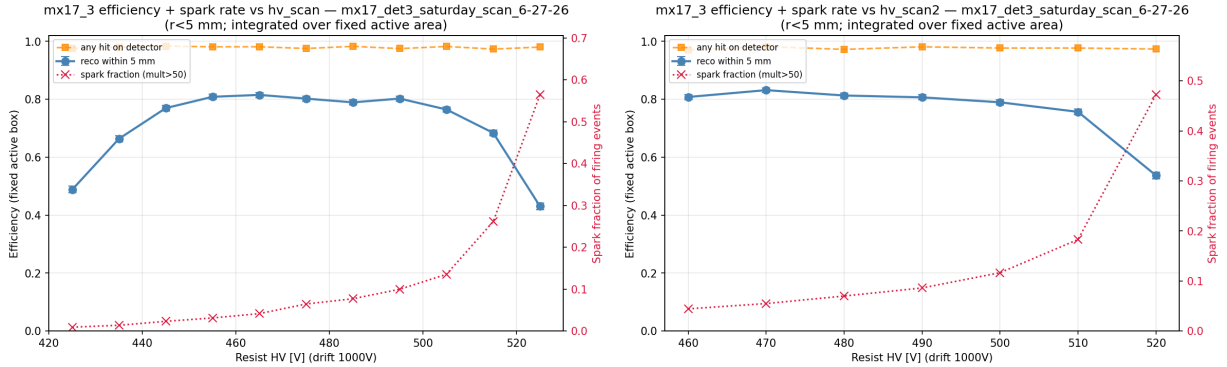


Figure 19: Resist-HV scans (left: 425–525 V; right: fine repeat 460–520 V). Efficiency (blue), any-hit response (orange), spark fraction (red crosses, right axis).

- **Turn-on** at 425 V (49 %) is complete by ~ 455 V;
- **plateau** 455–490 V at 80–83 % (best point 83.2 % at 470 V in the fine scan);
- **roll-off** above ~ 505 V is driven entirely by sparking: the spark fraction climbs from 4 % at 470 V to 56 % at 525 V while the any-hit response stays ≈ 98 % — the detector is not going silent, its events are being destroyed by discharges;
- the 490 V operating point of the long run sits on the plateau but already carries ~ 9 % spark fraction; 470–480 V gives the same efficiency at half the spark rate and is the recommended setting.

8 Summary and follow-ups

8.1 Summary

1. **Off-diagonal angle correlation — solved.** It is the additive charge-spreading bias $\tan \theta_{\text{det}} = \tan \theta_{\text{ref}} \pm w/z_{\text{vis}}$, $w/z_{\text{vis}} = 0.115$ ($w \approx 2.2$ mm): a detector-family constant. Not a rotation; not a bug. It also explains the $|\theta_{\text{det}}| \gtrsim 5^\circ$ exclusion gap and the horizontal arms at $\theta_{\text{ref}} \approx 0$.
2. **Drift velocity — measured.** The ridge fit turns the angle correlation into a precision v_d measurement: $v_d(1000 \text{ V}) = 28.1(7) \mu\text{m/ns}$, full curve in Table 2, 1–3 % per 15-minute point.
3. **Drift gap — 30 mm mechanical, 19.4 mm visible.** The clock-immune product $v_d T_{\text{sat}} = 19.4(2)$ mm is the depth at which drifting charge falls below threshold, not the electrode spacing: the arrival-time tail continues (exponentially suppressed) out to the 30 mm drift time at every field, and the strip amplitude decays with drift *depth* ($\lambda \approx 13 - 15$ mm, one universal curve for all fields) — charge from the far third of the gap is lost to attachment in the contaminated gas.
4. **Gas — contaminated with an electronegative polar impurity.** With the 30 mm field scale ($E = V/3.0 \text{ cm}$) the measured $v_d(E)$ and the attachment losses point together at air/moisture ingress into the Ar/iso 95/5 (Sec. 6); clean nominal gas and iso-rich mixing errors are excluded, and the gain behaviour rules out a wrong-bottle quencher swap.
5. **Operating point.** det3: resist 470–480 V (same efficiency as 490 V, half the sparks), drift $\geq 700 \text{ V}$ (readout-window constraint), 80 % reconstruction efficiency, 0.83/0.92 mm core resolution, 1.5–2° micro-TPC angular resolution for inclined tracks.

8.2 Follow-ups

1. The Saturday long run is *complete* at 141 min (it was stopped early when the p2 overnight run started at 01:33) — for more det3 statistics use the p2 run (53k rays) or take a new dedicated long run; the *reco-far* tail difference vs the p2 run can be probed with the existing data.
2. Gas system: fresh-gas re-scan after a long flush (one hour decides the contamination hypothesis — the velocity at fixed HV should jump *and* the visible column should extend towards 30 mm); humidity/O₂ probe on the exhaust; review flush time, flow rate and tubing permeability.
3. Mechanics: confirm the 30 mm drift-gap drawing (the tail endpoint gives 28–29 mm from data); confirm the DREAM sampling period (60 ns) to fix the absolute time scale.
4. Analysis: wire the ridge-fit v_d and the w/g angle correction into the production pipeline (`03_alignment_and_tpc.py`); revisit micro-TPC angular resolution after gas drying.
5. Longer readout window (or slower drift-field points avoided) if low-field drift data are wanted in the future.

Reproducibility. Run key `sat_det3` in `mx_june_cosmic_qa/qa_config.py`. Scripts: `13_tpc_angle_bias.py` (bias study), `14_drift_velocity_scan.py` (ridge fit vs HV), `15_drift_velocity_vs_magboltz.py` (gas comparison), `16_gap_from_pulse_width.py` and `17_gap_attachment_test.py` (gap tests), and `mm_drift_velocity_{table,candidates}.py`, `mm_attachment_{table,air_candidates}.py`, `mm_townsend_compare.py` in `garfield.sim/`

(Magboltz). Outputs live under `Analysis/mx17_det3_saturday_scan_6-27-26/` in the cosmic-bench data tree.